

PREDICTIVE ANALYTICS FOR HEALTHCARE: UTILIZING REGRESSION MODELS TO FORECAST PATIENT LENGTH OF STAY IN URBAN HOSPITALS

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By:

Name: Bidyabrat Goswami

Institute: Gauhati University

Institute Roll No.: UA-211-146-0023

Enrollment No.: IDRA-2026-181921

Under the Supervision of

Dr. Shaheena Salam

Through the platform:



India Data Research Academy

DECLARATION

I declare that this project report represents my own analytical work completed as part of the IDRA Data Science & AI Training Program. The analysis, modelling and interpretations presented here were developed from the assigned dataset, and external sources used in the project have been acknowledged.

ACKNOWLEDGEMENT

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ABSTRACT

This project investigates whether patient and admission characteristics can be used to predict hospital length of stay. The assigned dataset contains 61,680 hospital admissions and 28 original variables covering readmission history, demographic information, comorbidity indicators, laboratory measurements, facility information and admission/discharge dates. The workflow followed data understanding, quality checking, cleaning, feature engineering, exploratory data analysis, descriptive statistics and regression modelling. The raw data contained no missing values or duplicate rows; one physiologically invalid negative glucose value was treated as missing, while the top-coded readmission category '5+' was converted to 5. An 80:20 train-test split was used. Linear Regression was used as a transparent baseline and a Random Forest Regressor was used as the main nonlinear model. The Random Forest achieved an MAE of 0.49 days, RMSE of 0.72 days and R^2 of 0.905 on the test set, substantially outperforming the linear baseline. Readmission count was the strongest simple numerical association with length of stay, while facility and several comorbidity indicators also showed meaningful differences. The results support predictive analytics as a planning and decision-support aid, while recognising that the model does not establish causation and should not replace clinical judgement.

Keywords: healthcare analytics; length of stay; exploratory data analysis; regression; Random Forest; predictive analytics

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1. INTRODUCTION

1.1 Background

Hospital length of stay (LOS) is an important operational outcome because it affects bed availability, staffing, discharge planning and resource utilisation. A model that estimates LOS from information available around admission can help hospitals anticipate workload and organise resources. The IDRA capstone guide identifies patient characteristics, admission information, hospital operations and stay duration as the primary focus of Project 1.

1.2 Context of the Problem

The present dataset contains patient-level encounter information, including previous readmission count, gender, facility, comorbidity indicators and laboratory/vital measurements. The analytical task is regression because the target, lengthofstay, is a numerical duration measured in days. The project focuses on associations and prediction rather than causal inference.

1.3 Scope of the Study

The study covers data quality assessment, preprocessing, exploratory data analysis, descriptive statistics, feature engineering and supervised regression. The analysis does not attempt to diagnose patients, determine treatment, establish causal effects, or deploy the model in a live hospital environment.

2. PROBLEM STATEMENT AND MOTIVATION

The central problem is: Can patient and admission characteristics available at or around admission be used to predict the number of days a patient will remain in hospital? Accurate estimates could support capacity planning, staffing and discharge preparation. A major modelling requirement is avoiding information leakage; therefore, the post-admission discharge date is excluded from the predictor set.

3. OBJECTIVES

- Understand the structure, quality and distribution of the assigned hospital length-of-stay dataset.
- Identify important relationships between patient/admission characteristics and length of stay.
- Prepare the data using justified cleaning, encoding, scaling and feature-engineering decisions.
- Build a regression model to predict patient length of stay.
- Evaluate the model using MAE, MSE, RMSE and R^2 and assess evidence of overfitting.
- Translate the analytical results into evidence-based conclusions and practical recommendations.

4. RESEARCH QUESTIONS

1. What are the main characteristics and quality issues in the assigned dataset?

2. Which variables are most strongly associated with length of stay?
3. How does length of stay vary across readmission counts, facilities and selected comorbidity groups?
4. How accurately can a Random Forest model predict length of stay from admission-time information?
5. Does the final model show evidence of overfitting when training and test performance are compared?

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

5.1 Data Source

The dataset used in this capstone was the Project 1 CSV supplied through the IDRA LMS:

P_1_LengthOfStay.csv. The variable definitions were cross-checked against Microsoft's documentation for the hospital length-of-stay dataset schema, which describes fields such as readmission count, laboratory values, facility ID, visit date, discharge date and length of stay.

5.2 Dataset Size and Structure

The raw dataset contains 61,680 observations and 28 columns. The target variable is lengthofstay, ranging from 1 to 17 days, with mean 4.00, median 4.0 and standard deviation 2.36.

Measure	Value	Role	Notes
Observations	61,680	Dataset size	Hospital admissions
Original variables	28	Dataset size	Before derived features
Target	lengthofstay	Target	Regression outcome, days
Categorical predictors	gender, facid	Features	One-hot encoded
Readmission count	rcount → rcount_num	Feature	5+ coded as 5
Admission date	vdate → month/weekday	Feature	Raw date excluded
Discharge date	discharged	Excluded	Avoids target leakage

Table 1. Summary of the Dataset

5.3 Data Dictionary

The compact data dictionary below lists the main variables used to understand and model length of stay.

Variable	Meaning	Type	Role
eid	Unique admission identifier	Integer	Excluded
vdate	Hospital visit/admission date	Date string	Transformed
rcount	Readmissions within prior 180 days	Categorical	Feature
gender	Patient gender	Categorical	Feature
facid	Facility identifier	Categorical	Feature
dialysisrenalendstage	Renal disease/dialysis indicator	Binary	Feature
hematocrit	Average hematocrit measurement	Numeric	Feature
glucose	Average glucose	Numeric	Feature

	measurement		
secondarydiagnosisisonicd9	Secondary diagnosis indicator/count field	Numeric	Feature
discharged	Discharge date	Date string	Excluded to prevent leakage
lengthofstay	Length of hospital stay in days	Integer	Target

Table 2. Data Dictionary

6. DATA CLEANING

The raw data were checked for missing values, duplicate records, data-type consistency, invalid values and potential outliers. Cleaning decisions were kept conservative so that unusual but potentially genuine clinical observations were not removed without justification.

6.1 Missing Values

No missing values were present in the raw dataset. During validation, one negative glucose value was identified as physiologically invalid. Rather than deleting the entire record, the invalid glucose measurement was converted to missing and handled through median imputation within the modelling pipeline.

6.2 Duplicate Records

No duplicate rows were found in the raw dataset. The row count therefore remained 61,680 after cleaning.

6.3 Data Types and Inconsistent Values

The date fields were parsed from strings into datetime objects. The readmission-count field contained the category '5+', which was converted to the numeric value 5 to make the variable usable in regression. Binary condition indicators were retained as numeric 0/1 variables.

6.4 Outlier Handling

Several numerical variables contain extreme values. These were not automatically deleted because an extreme clinical measurement can represent a genuine patient state. The project therefore uses median imputation only for the invalid negative glucose value and retains other observations unless demonstrably invalid.

6.5 Before vs After Cleaning

Quality item	Before	After	Action
Rows	61,680	61,680	No row deletion
Columns	28	33	Five derived fields added
Missing values	0	1 in glucose	Invalid negative glucose converted to NaN
Duplicate rows	0	0	No action required
rcount '5+'	3,050	3,050	Mapped to 5; numeric conversion

Table 3. Data Quality Before and After Cleaning

7. DATA PREPROCESSING

The cleaned dataset was prepared for modelling using categorical encoding, numerical imputation and scaling, feature transformation and an 80:20 train-test split. Preprocessing objects were fitted inside the training pipeline to avoid test-set information leakage.

7.1 Categorical Encoding

Gender and facility ID were one-hot encoded. One-hot encoding avoids imposing an artificial numerical order on categorical labels.

7.2 Feature Scaling and Transformation

Numerical predictors were median-imputed and standardized within the training pipeline. Scaling is useful for the linear baseline and keeps preprocessing consistent across the modelling workflow. Tree-based Random Forest prediction is not dependent on feature scaling, but the common pipeline ensures a reproducible transformation process.

7.3 Train-Test Split

The data were divided into 80% training and 20% testing sets using `random_state=42`. The preprocessing objects were fitted through the training pipeline, preventing information from the test set from influencing learned transformations.

7.4 Final Feature Set

The final feature set contains readmission count, laboratory/vital variables, comorbidity indicators, secondary diagnosis information, admission month, admission weekday, gender and facility. `eid` was excluded as an identifier, and `discharged` was excluded because it is post-admission information directly related to the final length of stay.

8. EXPLORATORY DATA ANALYSIS

EDA was designed to answer specific questions about the target distribution, group differences and relationships rather than to add charts without analytical purpose.

8.1 Univariate Analysis

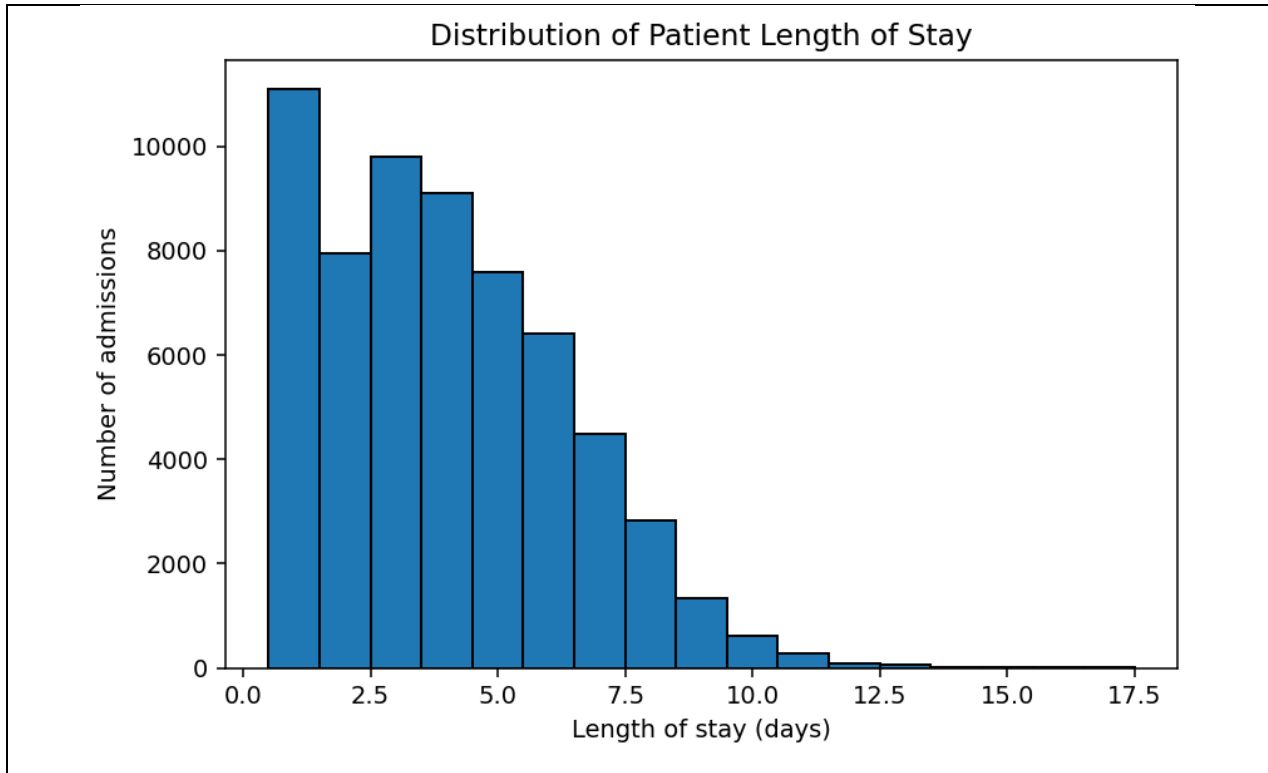


Figure 1. Distribution of Patient Length of Stay

Figure 1 shows that length of stay is concentrated around the middle of the 1–17 day range, with the mean and median both close to 4 days. The bounded integer target is suitable for regression, although predictions should be interpreted as estimates of expected stay duration rather than exact discharge dates.

8.2 Categorical Analysis

Readmission history and facility were examined as important categorical/grouping variables because both may capture differences in patient complexity or operational context.

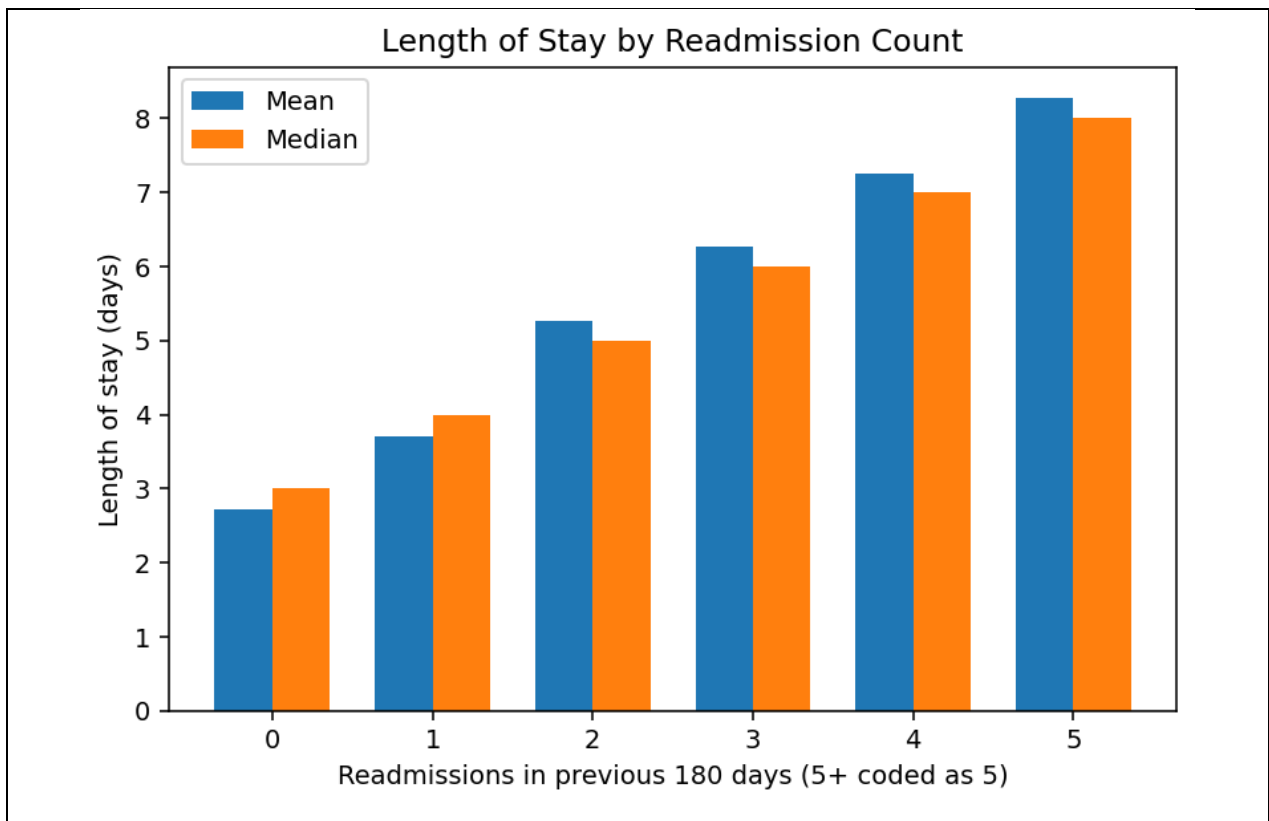


Figure 2. Length of Stay by Readmission Count

Figure 2 shows a strong monotonic pattern: mean length of stay rises from about 2.72 days for patients with no recent readmissions to about 8.29 days for the 5+ group. This is also reflected in the correlation between numeric readmission count and LOS ($r \approx 0.75$).

8.3 Bivariate / Relationship Analysis

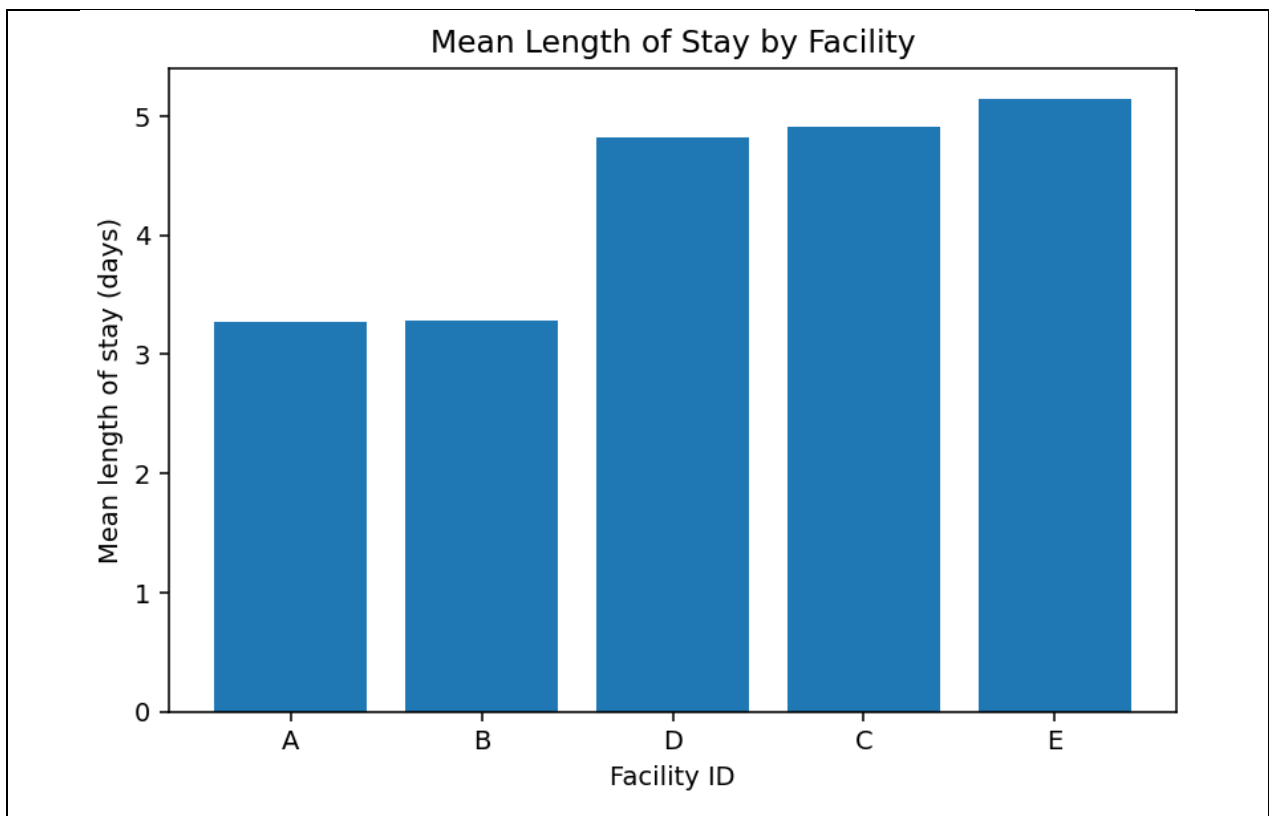


Figure 3. Mean Length of Stay by Facility

Figure 3 shows facility-level differences. Mean stay is approximately 3.28 days at A, 3.29 at B, 4.92 at C, 4.82 at D and 5.15 at E. These differences may reflect differences in case mix or facility operations; the analysis does not establish the cause.

Comorbidity comparisons also show that several binary condition indicators are associated with longer average stays. Dialysis/renal end-stage, psychotherapy, fibrosis/other conditions, hemo/blood disorder, malnutrition and pneumonia were among the largest observed group differences.

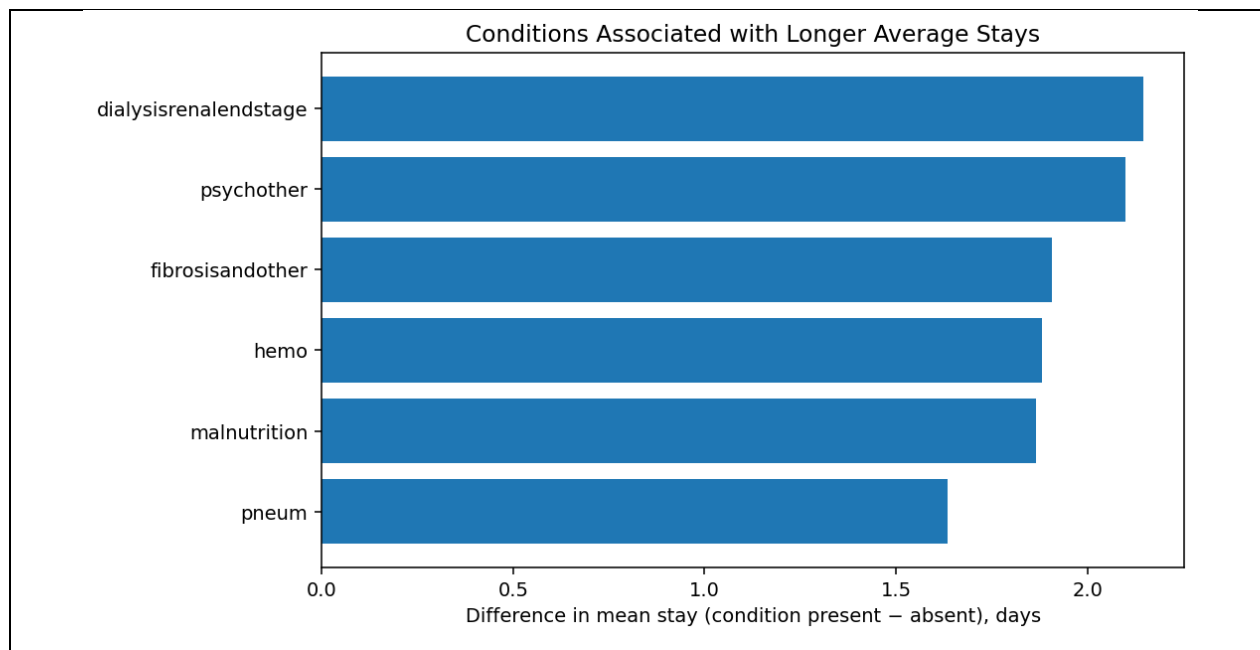


Figure 4. Conditions Associated with Longer Average Stays

Figure 4 highlights conditions with the largest difference in average LOS between patients with and without the indicator. Dialysis/renal end-stage shows the largest difference, followed by psychotherapy, fibrosis/other conditions, hemo/blood disorder, malnutrition and pneumonia. These are group-level associations, not causal effects.

8.4 Correlation Analysis

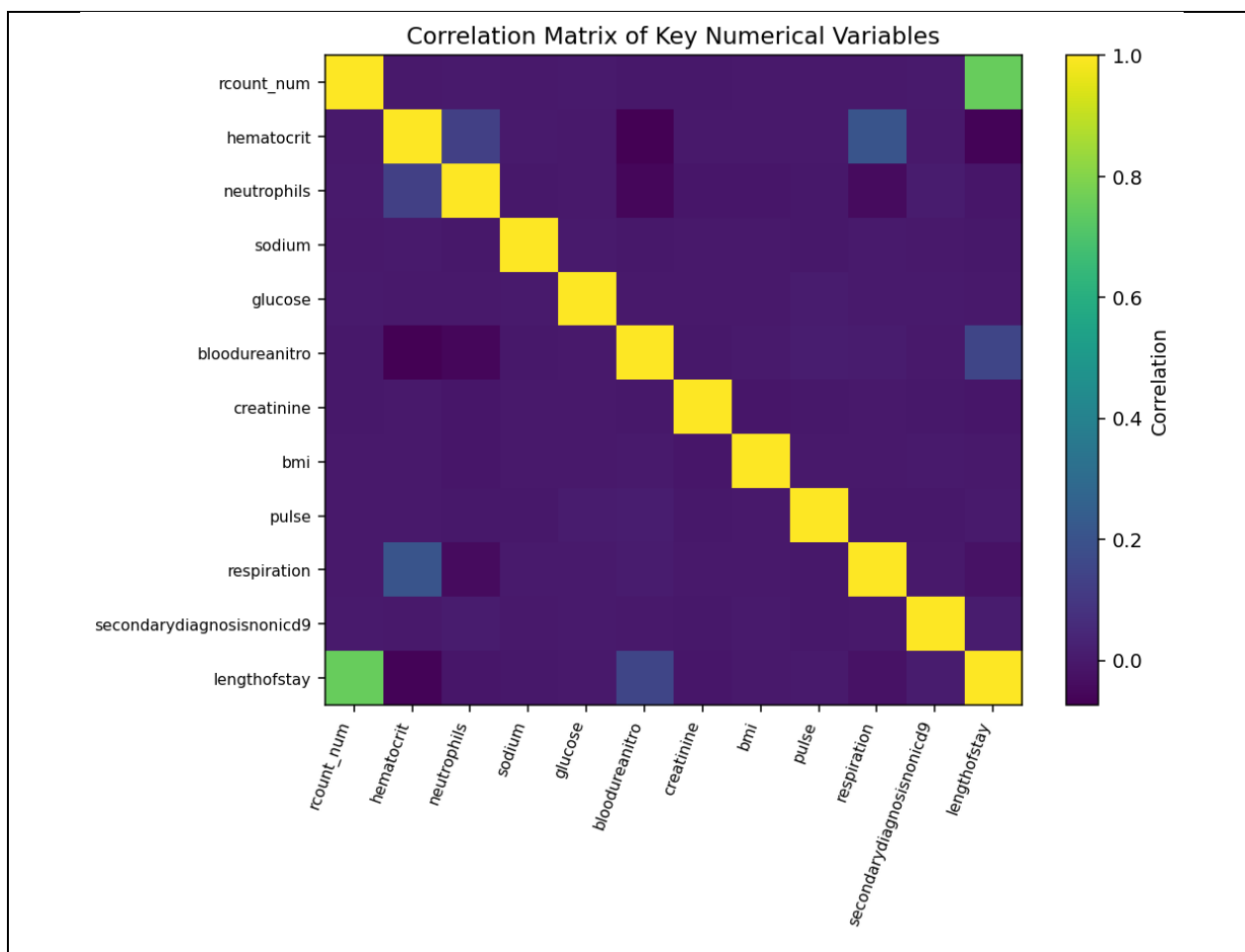


Figure 5. Correlation Heatmap of Numerical Variables

Figure 5 confirms that rcount_num has the strongest numerical relationship with LOS. Several comorbidity indicators are positively associated with longer stays, while most laboratory variables have weak linear correlations with LOS in this dataset. Weak linear correlation does not mean that a variable has no predictive value because nonlinear and interaction effects may still be captured by tree-based models.

8.5 Other Figures

The Random Forest feature-importance plot provides a model-based view of which variables contributed most strongly to prediction. Readmission count is the dominant feature, followed by facility indicators and several physiological/comorbidity variables.

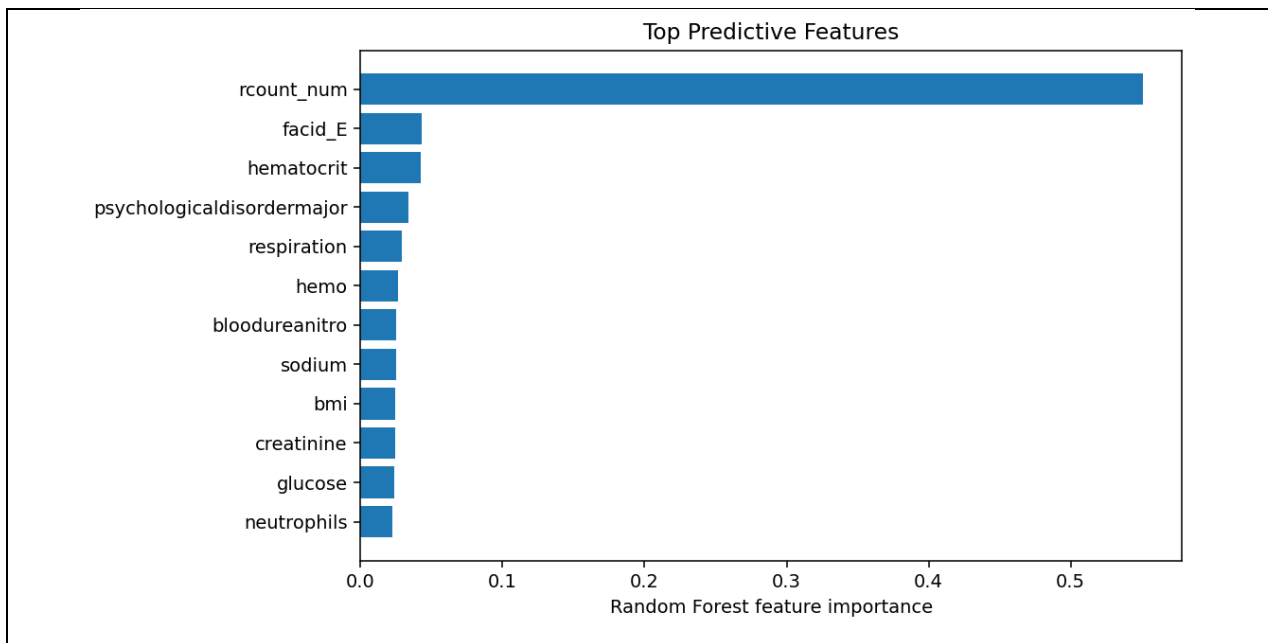


Figure 6. Random Forest Feature Importance

Feature importance should be interpreted as predictive importance, not causal influence.

9. STATISTICAL ANALYSIS

The descriptive statistics below summarise the target and key numerical predictors. The target has mean 4.00 days, median 4.00 days and standard deviation 2.36 days. Readmission count has a mean of 1.12 after converting 5+ to 5, and its relationship with LOS is much stronger than that of the continuous laboratory measurements.

Variable	Mean	Median	SD	Min–Max
lengthofstay	4.00	4.00	2.36	1.00–17.00
rcount_num	1.12	0.00	1.54	0.00–5.00
hematocrit	11.97	11.90	2.04	4.40–24.10
neutrophils	10.20	9.40	5.27	0.10–245.90
sodium	137.90	137.90	3.00	124.91–151.39
glucose	142.06	142.25	29.92	11.05–268.57
bloodureanitro	14.11	12.00	12.75	1.00–682.50
creatinine	1.10	1.10	0.20	0.22–2.04
bmi	29.80	29.80	2.00	22.04–38.94
pulse	73.42	73.00	11.65	21.00–126.00
respiration	6.49	6.50	0.57	0.20–10.00
secondarydiagnosisisonicd9	2.12	1.00	2.05	0.00–10.00

Table 4. Descriptive Statistics of Key Variables

Group summaries similarly show higher mean stays for several comorbidity indicators, supporting their inclusion as candidate predictive features.

10. FEATURE ENGINEERING

Three main transformations were used: rcount was converted from a mixed categorical field to numeric rcount_num; vdate was parsed and reduced to admission month and admission weekday; and the single

negative glucose value was converted to missing for safe imputation. These transformations preserve information that is plausibly available at admission without using post-admission discharge information.

Original	Engineered feature	Transformation	Purpose
rcount	rcount_num	5+ → 5; integer	Represent readmission history numerically
vdate	admission_month	Extract month	Capture seasonal/admission timing patterns
vdate	admission_weekday	Extract weekday	Capture day-of-week effects
glucose	validated glucose	Negative value → NaN	Allow safe median imputation

Table 5. Engineered Features

11. MODEL DEVELOPMENT

11.1 Problem Formulation

The task is supervised regression. The target is lengthofstay, measured in days. The predictors describe patient history, comorbidity indicators, laboratory/vital measurements, facility and admission timing.

11.2 Model Selection

Linear Regression was selected as a transparent baseline because it provides a simple reference for how much predictive performance can be achieved with approximately linear effects. Random Forest Regressor was selected as the primary model because the EDA indicates strong group differences and likely nonlinear relationships/interactions. The Random Forest also provides feature-importance estimates for interpretation.

11.3 Training the Model

The final Random Forest used 150 trees, minimum leaf size 2, square-root feature sampling and random_state=42. The model was trained on the 80% training split. All preprocessing was contained within the scikit-learn pipeline.

```
from sklearn.ensemble import RandomForestRegressor
model = RandomForestRegressor(n_estimators=150, min_samples_leaf=2,
                             max_features="sqrt", random_state=42, n_jobs=2)
model.fit(X_train, y_train)
```

12. MODEL EVALUATION

The trained models were evaluated on unseen test data using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and R-squared (R^2). Lower MAE/MSE/RMSE indicate smaller prediction errors, while higher R^2 indicates that a larger proportion of target variance is explained by the model.

12.1 Regression Evaluation

Model	MAE	MSE	RMSE	R^2
Linear Regression	0.8660	1.3038	1.1419	0.7624
Random Forest	0.4874	0.5185	0.7201	0.9055

Table 6. Regression Model Performance

The Random Forest reduces test MAE from 0.87 to 0.49 days and increases R^2 from 0.762 to 0.905. The final test RMSE is 0.72 days.

12.2 Classification Evaluation

Not applicable. This capstone is a regression task because lengthofstay is a numerical duration. Classification-specific metrics such as a confusion matrix, accuracy, precision, recall and F1-score are therefore not used.

12.3 Overfitting and Underfitting

Training performance for the Random Forest is $R^2=0.969$, RMSE=0.42 days and MAE=0.28 days, compared with test $R^2=0.905$, RMSE=0.72 days and MAE=0.49 days. The gap indicates some overfitting, which is expected for a flexible ensemble, but the test performance remains strong relative to the linear baseline.

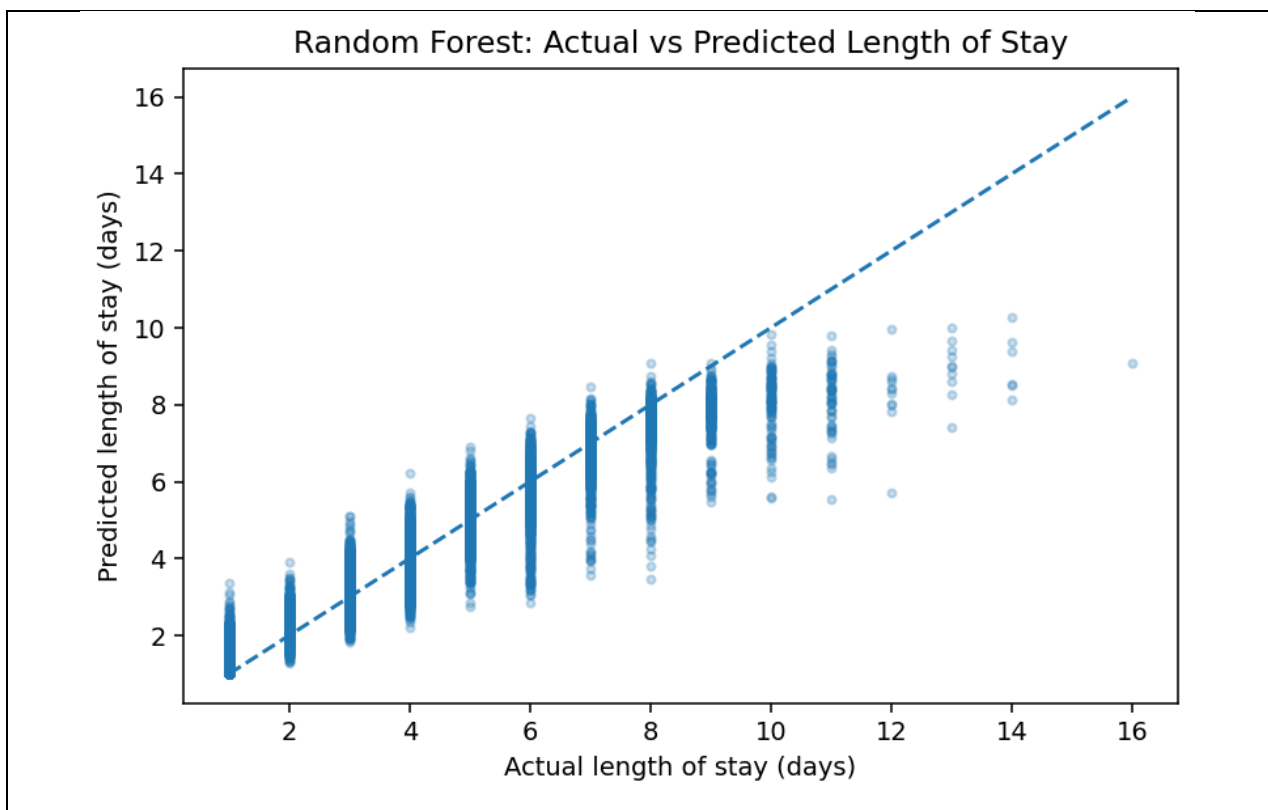


Figure 7. Random Forest: Actual vs Predicted Length of Stay

Figure 7 compares actual and predicted LOS. Points close to the diagonal represent accurate predictions. The model captures the overall pattern well, although exact prediction becomes more difficult for individual cases.

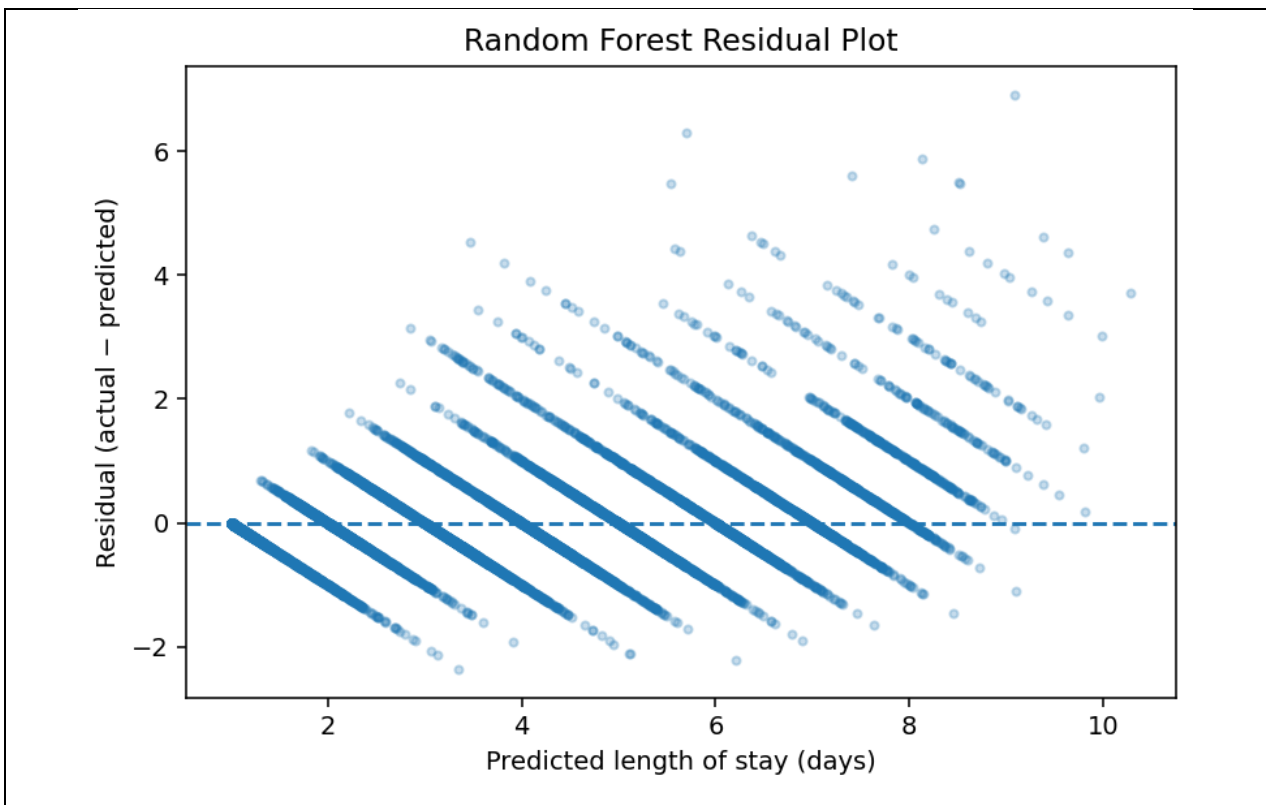


Figure 8. Random Forest Residuals

The residuals show remaining prediction error around the zero line, particularly for individual admissions. This is consistent with the model being a useful estimate rather than a deterministic discharge rule.

13. RESULTS AND FINDINGS

6. 1. Readmission history is the strongest simple predictor: rcount_num correlates with LOS at approximately 0.75, and mean LOS increases from 2.72 days at zero readmissions to 8.29 days in the 5+ group.
7. 2. Facility is an important categorical factor, with average LOS ranging from about 3.28 days at facility A to about 5.15 days at facility E.
8. 3. Several comorbidity indicators are associated with materially longer stays; dialysis/renal end-stage has the largest mean difference among the analysed binary conditions.
9. 4. The Random Forest substantially outperforms Linear Regression, achieving MAE=0.49 days and $R^2=0.905$ on unseen test data.
10. 5. The model retains a train-test performance gap, so predictions should be monitored and validated on new hospital populations before operational use.

14. DISCUSSION

The analysis suggests that previous hospital utilisation and patient complexity are more informative for LOS prediction than many individual laboratory measurements in this dataset. The strong readmission pattern may reflect underlying complexity, recurrent illness or other unmeasured factors. Facility differences may reflect case mix, service availability or operational practices, but the dataset does not support attributing the difference to any one cause. The Random Forest's substantially better performance than the linear baseline indicates that LOS relationships are not adequately represented by a purely linear model. Nevertheless, the residuals and train-test gap show that important variation remains unexplained. A practical interpretation is that the model can provide an approximate planning estimate while still requiring human review.

15. LIMITATIONS

- The dataset represents a particular hospital/encounter population and may not generalise to other hospitals or regions.
- The dataset does not contain every clinically or operationally relevant determinant of discharge timing, so unexplained variation remains.
- The data include unusual laboratory values; only the clearly invalid negative glucose value was corrected, while other extreme observations were retained.
- The model is predictive and observational; correlations and feature importance do not establish causal effects.
- The final evaluation uses one random train-test split rather than external prospective validation.

16. CONCLUSION

This capstone demonstrates a complete data-science workflow for forecasting patient length of stay. The assigned dataset was inspected, cleaned, transformed and analysed using descriptive statistics and targeted visualisation. Readmission count emerged as the strongest simple association with LOS, while facility and several comorbidity indicators also showed meaningful differences. A Random Forest Regressor achieved an MAE of 0.49 days, RMSE of 0.72 days and R^2 of 0.905 on the test set, outperforming the linear baseline. The model therefore shows useful predictive potential for planning support, subject to external validation and responsible use.

17. RECOMMENDATIONS AND FUTURE WORK

17.1 Recommendations

- Use readmission history as an important input to early LOS risk stratification because it shows the strongest observed association with the target.
- Consider facility-specific planning when estimating bed demand because average LOS differs substantially across facility groups.
- Give additional attention to patients with major comorbidity indicators associated with longer stays, while treating these variables as risk markers rather than causes.
- Use the model as a decision-support estimate alongside clinical and operational judgement rather than as an automatic discharge decision rule.
- Before operational deployment, validate the model on a separate hospital population and monitor performance over time.

17.2 Future Work

Future work could include external validation, time-aware validation, additional clinical variables, more detailed diagnosis/procedure information, calibration of prediction intervals, comparison with additional tree-based models, and prospective monitoring of prediction error.

18. REFERENCES

India Data Research Academy. (2026). Capstone Project Guide 2026. Data Science & AI Training Program.

India Data Research Academy. (2026). P_1_LengthOfStay.csv. Project 1 dataset supplied through the IDRA LMS.

Microsoft. Predicting Hospital Length of Stay — Input Data documentation. Microsoft Machine Learning Services. https://microsoft.github.io/r-server-hospital-length-of-stay/input_data.html

APPENDIX A. CODE

The following key code excerpts support reproducibility. The complete executable notebook is submitted separately with the project files.

A.1 Data Loading and Inspection

```
df = pd.read_csv("P_1_LengthOfStay.csv")
print(df.shape)
print(df.isna().sum())
print(df.duplicated().sum())
```

A.2 Data Cleaning

```
clean["rcount_num"] = clean["rcount"].replace({"5+": 5}).astype(int)
clean["admission_date"] = pd.to_datetime(clean["vdate"])
clean.loc[clean["glucose"] < 0, "glucose"] = np.nan
```

A.3 Preprocessing

```
ColumnTransformer([
    ("num", Pipeline([("imputer", SimpleImputer(strategy="median")),
                      ("scaler", StandardScaler())]), numeric),
    ("cat", OneHotEncoder(handle_unknown="ignore"), categorical)
])
```

A.4 EDA and Visualization

Matplotlib visualisations were used to examine LOS distribution, readmission and facility differences, comorbidity comparisons, correlations, predictions and residuals.

A.5 Model Training and Evaluation

```
RandomForestRegressor(n_estimators=150, min_samples_leaf=2,
                      random_state=42, n_jobs=2, max_features="sqrt")
MAE = mean_absolute_error(y_test, pred)
RMSE = mean_squared_error(y_test, pred, squared=False)
R2 = r2_score(y_test, pred)
```

APPENDIX B. ADDITIONAL FIGURES AND TABLES

The appendix provides supplementary model outputs that support interpretation but are not essential to the main narrative.

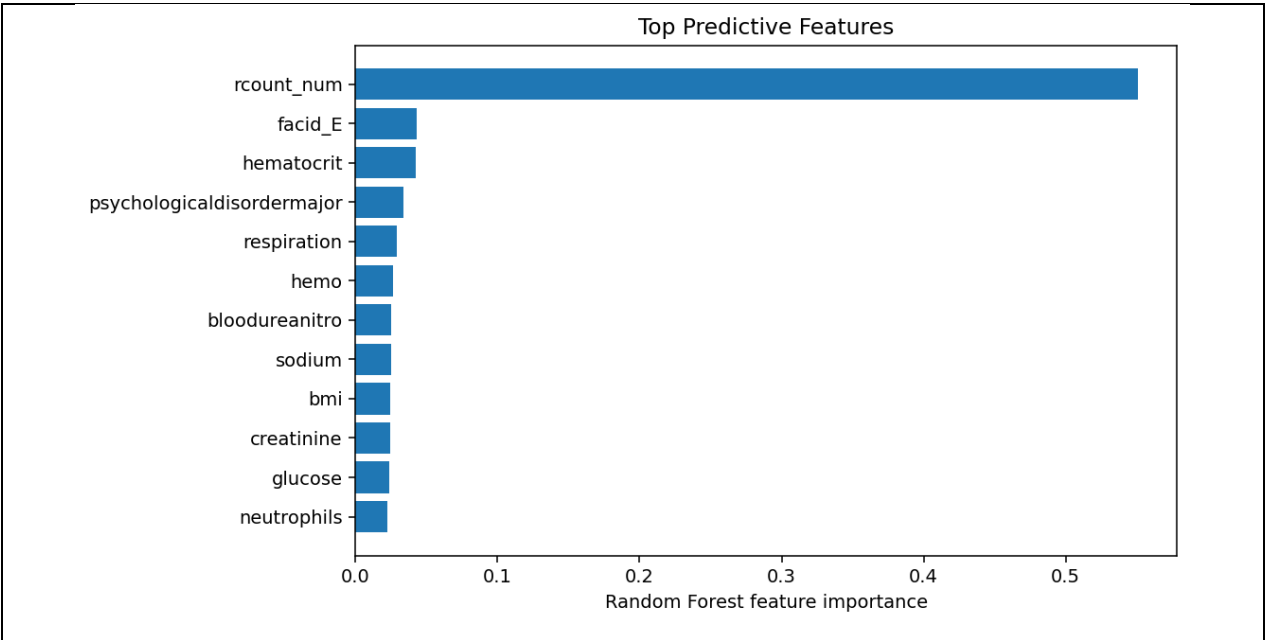


Figure 9. Random Forest Feature Importance (Additional View)

Metric	Linear Regression	Random Forest	Interpretation
MAE	0.8660	0.4874	Lower is better
RMSE	1.1419	0.7201	Lower is better
R ²	0.7624	0.9055	Higher is better
Test set	20%	20%	Unseen evaluation data

Table 7. Additional Model Results